• Mini Project Report on Customer Segmentation using K-Means Clustering

• Aim:

To perform customer segmentation using K-Means clustering and visualize groups based on income and spending behavior using R.

• Scenario / Background:

- A retail store collects data about its customers including their annual income and spending habits.

  The management wants to:
  - Identify different types of customers based on purchasing behavior.
  - Group customers into clusters for targeted marketing strategies.
  - Understand high-value and low-value customer segments.
  - Improve business decisions by analyzing customer patterns

• Sample Dataset:

Customer ID	Income	Spending Score
1	15000	20
2	18000	25
3	25000	30
4	30000	35
5	45000	50
6	50000	55
7	65000	65
8	70000	70

- **R Program (with line-by-line comments):**

```
# Create dataset (as given)
 customer <- data.frame(
  CustomerID = 1:8,
  Income = c(15000, 18000, 25000, 30000, 45000, 50000, 65000, 70000),
  Spending = c(20, 25, 30, 35, 50, 55, 65, 70)
)
# Display dataset
print(customer)
# Select required columns
data <- customer[, c("Income", "Spending")]
# Normalize the data
data_scaled <- scale(data)
# Apply K-Means clustering
set.seed(123)
kmeans_result <- kmeans(data_scaled, centers = 3)
# Add cluster labels to dataset
customer$Cluster <- kmeans_result$cluster
# Display result
print(customer)
# Plot clusters
plot(customer$Income, customer$Spending,
     col = customer$Cluster,
     pch = 19,
     main = "Customer Segmentation using K-Means",  xlab = "Income",
```

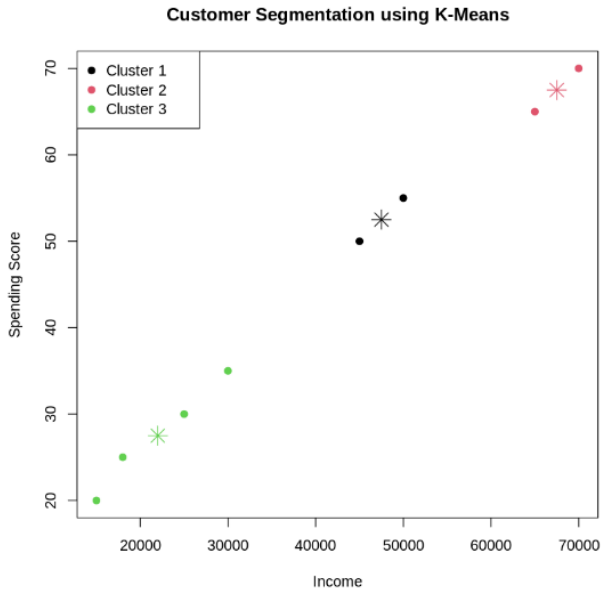
```
  ylab = "Spending Score")
# Add cluster centers
centers <- kmeans_result$centers
centers_original <- t(apply(centers, 1, function(x)
  x * attr(data_scaled, "scaled:scale") + attr(data_scaled, "scaled:center")))
points(centers_original,
       col = 1:3,
       pch = 8,
       cex = 2)
# Add legend
legend("topleft",
       legend = c("Cluster 1", "Cluster 2", "Cluster 3"),
       col = 1:3,
       pch = 19)
```

- **Output :**

Customer ID	Income	Spending
1	15000	20
2	18000	25
3	25000	30
4	30000	35
5	45000	50
6	50000	55
7	65000	60
8	70000	70

Customer ID	Income	Spending	Cluster
1	15000	20	3
2	18000	25	3
3	25000	30	3
4	30000	35	3
5	45000	50	1
6	50000	55	1
7	65000	65	2
8	70000	70	2

- **Graph:**



- **Overall Insight:**

Clustering reveals distinct customer profiles that the business can use for strategy:

- **High-Value Segment:** High-income and high-spending customers (Cluster 2).
- **Potential/Balanced Segment:** Moderate income and spending (Cluster 1).
- **Low-Value Segment:** Low-income and low-spending customers (Cluster 3).

- **Observation:**

1. The dataset shows clear variation in customer income and spending scores, indicating that customers behave differently despite similar earnings.
2. Clustering reveals distinct customer segments such as:
3. High-income, high-spending customers (premium or high-value customers)
4. High-income, low-spending customers (potential customers who can be targeted to increase spending)
5. Low-income, high-spending customers (impulsive or frequent buyers)
6. Low-income, low-spending customers (low-value segment)
7. Customers with similar purchasing behavior are grouped together, making it easier to understand patterns and preferences.
8. The clustering helps identify target groups for marketing campaigns, allowing personalized offers and promotions.

9. High-value customers can be prioritized for loyalty programs, while low-value customers can be targeted with strategies to improve engagement.

- **Submitted by:**

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Topic: **Mini Project Report on Customer Segmentation using K-Means Clustering**

Data Science Using R Programming Lab